NETWORK TOPOLOGIES

Course: Diploma in Cyber Security

Topic: Network Topologies

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Introduction

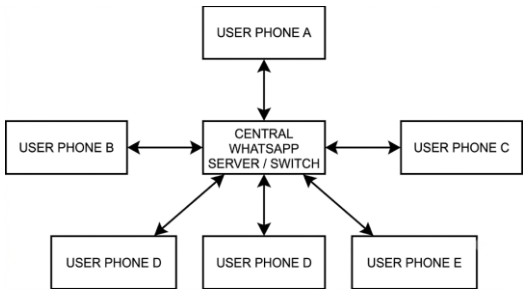
A network topology tells us how computers and other devices are connected in a network. Different topologies have different advantages, disadvantages and uses. In this assignment, I have studied star, bus, ring and mesh topologies and used simple daily-life examples to understand them.

Task 1: Draw and Label Network Topologies

For this task, only the three figures below need to be added. You can draw them on paper and take a clear photo, or make them using any drawing tool.

1. Star Topology

All devices are connected to one central switch or hub. It is easy to manage and if one device or cable has a problem, the other devices can normally keep working.



FIGURE

Real-world example: WhatsApp messaging infrastructure

Why: A central-server setup can handle messages and media from many users. This makes communication easier to manage than direct connections between every phone.

2. Bus Topology

All devices share one main backbone cable. Data travels through the common cable to reach the required device.

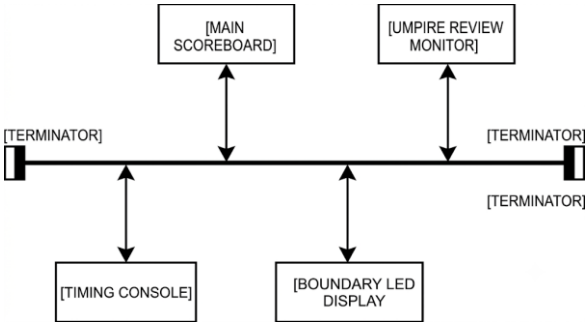


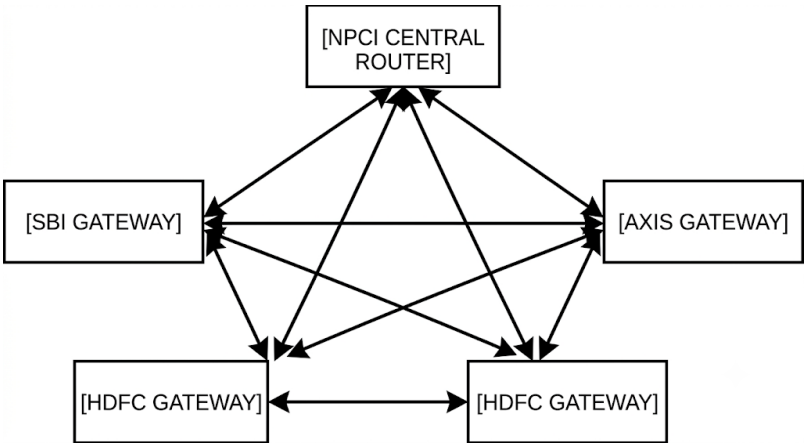
FIGURE:

Real-world example: IPL in-stadium scoreboards and timers (learning example)

Why: A shared communication line can carry simple status or timing updates in a small linear setup. The main benefit is a simple and low-cost layout.

3. Mesh Topology (Full Mesh)

Every device has a direct connection to every other device. This gives more than one possible path, so the network can continue working if one link fails.



FIGURE

Real-world example: UPI payment settlement nodes (learning example)

Why: A highly reliable system can use multiple network paths between important nodes. If one path has a problem, another path can be used.

Task 2: Comparison of Network Topologies

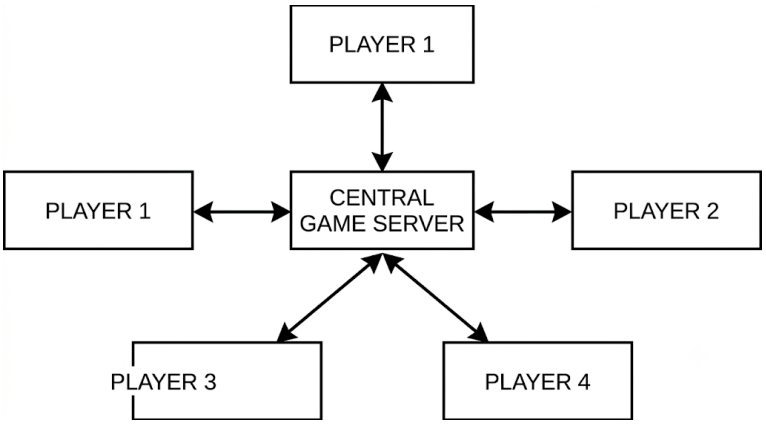
Topology	Pros (4)	Cons (4)	Best Use Case	App Example
Star	1. Easy to manage 2. Easy to add devices 3. Faults are easier to find 4. One device failure usually does not stop others	1. Needs a central switch/hub 2. Central device failure can stop the network 3. Uses more cable 4. Setup cost can be higher	Home, college or office LAN	Instagram
Bus	1. Simple design 2. Less cable needed 3. Low setup cost 4. Easy for a small basic network	1. Backbone failure can affect all devices 2. Performance can drop with more devices 3. Faults can be harder to find 4. Not good for large expansion	Small or temporary basic networks	Zomato
Ring	1. Organised data flow 2. Predictable path 3. Each device has a fixed place 4. Can work well in controlled networks	1. A break can affect communication 2. Adding devices is harder 3. Troubleshooting can take time 4. Changes may disturb the network	Special-purpose or controlled networks	Paytm
Mesh	1. Very reliable 2. Multiple paths available 3. Good fault tolerance 4. One link failure may not stop communication	1. Expensive 2. More cables/connections needed 3. Setup is complex 4. Maintenance is harder	High-reliability or critical networks	Large online services (learning example)

Note: Apps such as Instagram, Zomato and Paytm use a mix of networking technologies. The examples here are only for understanding topology concepts.

Task 3: Topology for a Multiplayer Mobile Game

Game example: A new multiplayer mobile game like PUBG or Free Fire.

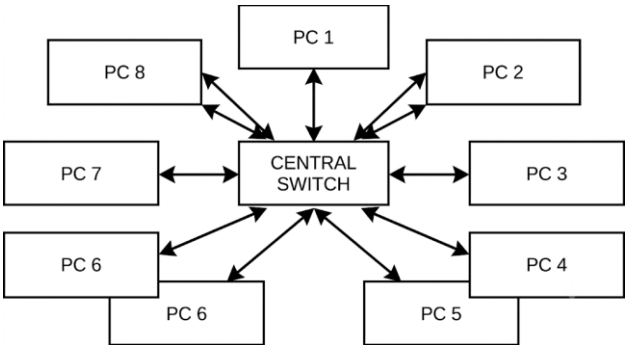
I would choose a star-like client-server topology for the game. Players connect to a central game server instead of directly connecting to every player. This can give good speed because the server manages the game data. It is also easier to control and troubleshoot, but server and network costs can be higher.



FIGURE

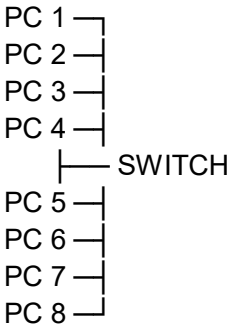
Task 4: Network for a Small Warehouse

For a small Flipkart-style warehouse with 8 computers, I would choose star topology. All 8 computers can connect to one central switch. It is easy to troubleshoot because each computer has its own connection, and more computers can be added later.



FIGURE

Quick sketch idea:



Conclusion

From this assignment, I learned that network topology decides how devices are connected. Star is easy to manage, bus is simple, ring gives an organised path, and mesh gives better reliability because it has multiple links. The best topology depends on cost, speed, reliability, troubleshooting and future expansion.